Vedant Desai

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EDUCATION

University of Michigan Expected May 2028
B.S. in Computer Science and Economics Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

Michigan Data Consulting (MDC) Jan 2026 – May 2026
Data Engineer — Michigan Campaign Finance Network Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at ∼2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating ∼800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.
- Architected a deterministic aggregation engine that parses those exports, normalizes irregular contribution amounts, and ranks PACs by total funding volume so researchers stop rebuilding spreadsheets to surface top spenders.

Vylet May 2026 – Present | vyletdata.com
Founder — Automated lead-sourcing platform for PE/search-fund; live product generating $1,500 MRR across three paying clients

- Launched Vylet, automating a ∼30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Architected a custom asyncpg Data Access Layer storing Gemini embeddings alongside source records; engineered injection-safe SQL timestamp validation that detects stale entries and triggers automatic re-scrapes, keeping the lead database fresh without manual intervention.

Lyndbrook Capital Feb 2026 – Apr 2026
Data Engineering Consultant Remote

- Automated off-market deal sourcing by aggregating 2,500+ legal entities from EPA compliance databases and cross-referencing Google Maps API data to surface unlisted acquisition targets, eliminating 15 hours of manual prospecting per week for the fund's Principal.
- Contracted pre-LOI to build acquisition intelligence for a boutique search fund targeting water utility operators; aggregated EPA ECHO and MassGIS regulatory data into a unified PWSID entity database and delivered 800+ validated Day-1 acquisition targets within the engagement window.

PROJECTS

SignalWeaver | *Python, FastAPI, PostgreSQL, pgvector, Docker*

- Containerized the stack with Docker Compose (API + Postgres/pgvector + nginx frontend) and a GitHub Actions CI pipeline (frontend build, pytest, API image build on main).

Granular Synthesizer Plugin | *C++, JUCE, VST3, AU* github.com/Verdent06/granular-synth

- Configured CMake to compile VST3 and AU bundles from a single JUCE codebase — targeting a macOS universal binary (arm64 + x86_64) for native Apple Silicon and Intel — and audited every processBlock() path against a real-time safety checklist confirming zero heap allocations and zero lock acquisitions before release builds.
- Enforced the audio thread's zero-allocation constraint by pre-allocating a MemoryPool<Grain, 64> slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so processBlock() never calls new or delete after prepareToPlay() completes.

TECHNICAL SKILLS

Languages .... Python, TypeScript, C++, SQL, HTML/CSS
Frameworks .... Flask, FastAPI, JUCE
Databases .... PostgreSQL, Redis, pgvector
Cloud & Infrastructure .... AWS (EC2, S3), Docker, Celery, Git, GitHub Actions